

Advanced (Habanero)

Q1. Solve for x in the equation $2^x = 8$.

- (A) $x = 2$
- (B) $x = 3$
- (C) $x = 4$
- (D) $x = 5$
- (E) $x = 6$

Q2. Solve for y in the equation $3^y = 27$.

- (A) $y = 1$
- (B) $y = 2$
- (C) $y = 3$
- (D) $y = 4$
- (E) $y = 5$

Q3. Solve for z in the equation $4^z = 16$.

- (A) $z = 1$
- (B) $z = 2$
- (C) $z = 3$
- (D) $z = 4$
- (E) $z = 5$

Q4. Solve for x in the equation $2^{x+1} = 12$.

- (A) $x = 1$
- (B) $x = 2$
- (C) $x = 3$
- (D) $x = 4$
- (E) $x = 5$

Q5. Solve for y in the equation $5^{y-1} = 25$.

- (A) $y = 1$
- (B) $y = 2$
- (C) $y = 3$
- (D) $y = 4$
- (E) $y = 5$

Q6. Solve for z in the equation $2^{z-2} = 8$.

- (A) $z = 1$
- (B) $z = 2$
- (C) $z = 3$
- (D) $z = 4$
- (E) $z = 5$

Q7. Solve for x in the equation $3^{x+2} = 81$.

- (A) $x = 1$
- (B) $x = 2$
- (C) $x = 3$
- (D) $x = 4$
- (E) $x = 5$

Q8. Solve for y in the equation $4^{y-1} = 4$.

- (A) $y = 0$
- (B) $y = 1$
- (C) $y = 2$
- (D) $y = 3$
- (E) $y = 4$

Open Ended Questions (FRQ)

Q9. Solve for x in the equation $2^x + 3^x = 35$.

Q10. Solve for y in the equation $y^2 = 2^y$.

Chef's Tips

- Read each question carefully before answering.
- Use a pencil to write your answers.
- Show all work for free response questions.